

Regions of Interest (ROI) Analysis for Upper Limbs EEG Neuroimaging Schemes

Khin Pa Pa Aung, Khin Htar Nwe

University of Information Technology, Yangon, Myanmar

khinpapaaung@uit.edu.mm, khinhartnwe@uit.edu.mm

Abstract

The rapid advancement in machine learning and digital signal processing is the reason for many researchers to develop brain-computer interface (BCI) systems. Recently, Electroencephalography (EEG) is the most widely used brain signals for many BCI systems because of recent EEG devices. These devices are easy to use, low costs, versatile and portable. Most popular EEG applications are controlling for artificial devices and virtual objects, rehabilitation for disabled persons, and computer displays in a real-time system. Since EEG does not collect the activity of single neurons, it detects the signals when populations of neurons are active at the same time and it made volume conduction problem. Thus the event-specific cortex regions and robust features for high classification accuracy are still challenges. This paper analyzed EEG signals over source-based (ROI) direction and the SVM classifier is applied to measure the accuracy of the proposed ROI. This work combined preprocessing, segmentation, Source Imaging, ROI analysis, feature extraction, and classification. Defined ROI is mainly focused on upper limb Motor Imagery EEG signals. The classification results showed that the defined ROI is reliable for MI EEG data.

Key Words- Brain-Computer Interface (BCI), electroencephalography (EEG), sensor-based, source-based, regions of interest (ROI)

1. Introduction

Over the past few years, many studies have been done for neurology to diagnose sleep disorders, epilepsy, mental and brain neurodegenerative diseases, etc. Nowadays, neurology is not only for clinical purposes, but it also becomes the attractive technology for many interactive applications; by translating brain signals into a command, and then the system will automatically perform a specific function from the acquired input.

The most common brain-reading technologies are Magnetoencephalography (MEG), Functional Magnetic Resonance Imaging (fMRI) and Electroencephalography (EEG). These technologies may be a non-invasive or invasive method. Noninvasive means that electrodes are

placed over the scalp and no need for a surgeon while the invasive method was used in other neuroimaging technology.

One of the main objectives of neuroscience and neurotechnology is to bring the BCI system to non-invasive applications with high accuracy. In 1929, German Psychiatrist Hans Berger discovered brain activity from the closed skull and invented the term “electroencephalogram” (EEG). In 1957, Gray Walter recorded brain signals by noninvasively, with large numbers of electrodes and visualized brain activity with the topography. He showed that brain rhythms change according to the mental task demanded. From that time, EEG is one of the most attractive brain data for BCI technology. Besides, recent portable EEG devices make EEG as the most attractive way for BCI. It is easy to use, low costs, versatile and portable.

EEG based BCI system used brain patterns such as P300, Steady-State Visual Evoked Potentials (SSVEPs), Motion-Related Cortical Potentials (MRCP), and Motor Imagery (MI). MI-based BCI is more suitable for reasonable applications because of its spontaneous data by individual freedom without required any external responsiveness, such as hand-free gaming, rehabilitation, automatic wheelchairs, etc. Figure 1 shows an EEG based BCI system.

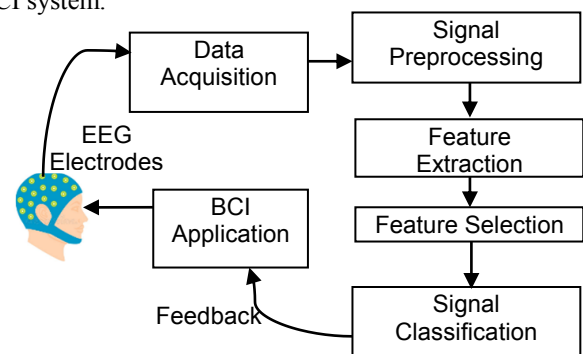


Figure 1: EEG based BCI system

Since a specific EEG signal cannot get from a single neuron, it was collected from the populations of neurons are activated at the same time through biological tissue towards measured sensors. It made a volume conduction

problem and thus still certain challenges in the EEG classification system that includes: localization of tasks specific brain cortical area and robust discriminated features between events/tasks. A lot of previous research has been done to overcome these challenges, by the methods of spatial filtering, windowing, and time combination approaches.

EEG signal analysis approaches can be classified into two main groups: source-based and sensor-based approaches. This paper focused on the source-based approach and the goal is to define robust ROI for MI EEG signals. By using Desikan-Killiany (DK) atlas, this work could reduce the signal complexity and extract the reasonable regions for a fast and reliable system. Four main steps for the MI-based EEG classification system are data acquisition, data preprocessing, feature extraction, and signals classification.

The rest of the paper is organized as follows: a brief description of Related Works, Proposed Materials and Methods, Results and Discussions, and Conclusions with Future research directions are elucidated in Section 2, 3, 4, and 5 respectively.

2. Related Works

Lots of previous studies have been done for the MI EEG classification system and primarily for the disabled person, hand-free gaming, and rehabilitation. Changes of brain rhythms depend on the specific stimuli of EEG signal modalities and task modalities such as imagine a specific task and cognitive task, and these rhythms come from certain cortical areas. Brain rhythms can be generally grouped by frequency bands such as delta (2-4 Hz), theta (5-7 Hz), alpha (8-12 Hz), beta (15-29 Hz), and gamma(>30 Hz) [1],[2], [3] and [7].

N.Korhan, Z. Dokur, and T.Olmez [2] 2019, studied over 4 different motor imagery tasks from three participants by combining CSP spatial filter and CNN classifier. The purpose of this studied was for high classification accuracy and 93.75% accuracy have been achieved with combined CSP and CNN.

K.Mahjoory, V.Nikulim, L Botrel, K.Linkenkaer, M.Fato, and S.Haufe [3] 2017, explained that the EEG inverse problem (source model) does not have a unique solution; it depends on forward and inverses modeling parameters. This study was used the resting state EEG recording within two studies was analyzed by using three software implementation. Finally, they encouraged that the verification of the obtained results using more than one source imaging procedures is essential.

In the study [4] 2016, M.Xing and R.Tadayonnejad proved that the EEG based connectivity reflects the cognitive load. This work was mainly for anxiety disorders and the weighted phase lag index (WPLI) was

used for connectivity matrices. The clustering coefficient and characteristic path length were computed to characterize this complex connectivity. Finally, the author concluded that the accurate neuroanatomical mapping will only possible with source mapping and this step was not covered in their work.

3. Proposed Materials and Methods

3.1. Dataset

A lot of EEG datasets can be freely available for different aims of BCI applications. EEG data can be recorded by two types: normal collected data and events related data.

This work used events related data, published by K. Murat, K. Mustafa, O. Erkan, Y. Hilmi and M. Yuriy [5]. Their main purpose is to overcome one of the main challenges in EEG BCI technology: “lack of large and uniform datasets” for the EEG researchers. To record EEG brain events, they used EEG-1200 station and 19 channels EEG cap, plus two ground leads (A1 and A2) and one bipolar lead (X3) for data synchronization. Thus a total of 22 channels ('Fp1' 'Fp2' 'F3' 'F4' 'C3' 'C4' 'P3' 'P4' 'O1' 'O2' 'A1' 'A2' 'F7' 'F8' 'T3' 'T4' 'T5' 'T6' 'Fz' 'Cz' 'Pz' 'X5') with international 10/20 standard system were applied to collect the EEG motor imagery (MI) dataset.

Among these 22 channels, this paper focused only on 19 channels. Three channels A1, A2, and X3 are excluded for this work that does not contain actual EEG data.

This dataset contained a total of 75 recordings (each recording took an average of 45 minutes) from 13 subjects and all the subjects are healthy students between the ages of 20 and 35 (8 males and 5 females). They performed “four interaction paradigms” named CLA, HaLT, FreeForm, and 5F paradigms respectively. The differences among these paradigms are “the number of imagery tasks” that needs to be performed by the subjects as displayed in eGUI.

This paper focused only on the CLA paradigm and which includes only 3 mental imagery tasks: left hand, right hand, and passive/neutral tasks. Each of these tasks was marked by numerical codes 1, 2, 3 respectively.

3.2. Data Preprocessing

3.2.1. Filtering. This paper mainly focused on ROI analysis. No custom filtering was applied to the raw EEG signal. A band-pass filter of 0.53-70Hz was already presented in all recordings and recorded at a 200 Hz sampling rate. Additionally, a 50 Hz notch filter was also presented in the EEG-1200 system to reduce electrical grid interference when collecting the raw data. These filters are hardware filters by the EEG-1200 system [5].

Figure 2 shows the proposed system design for this paper.

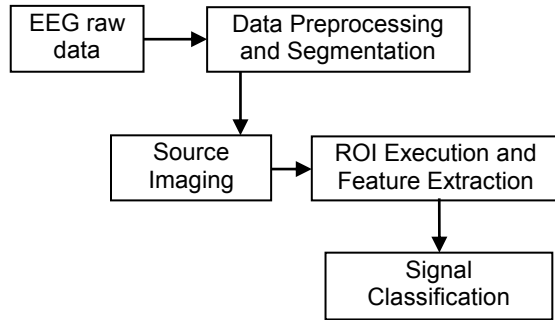


Figure 2: Proposed System Design

3.2.2. Data Segmentation. In this work, 14 EEG raw recording sessions from 6 different subjects (“each subject was named A, B, C, D, E, and F when recording”) are used. Each EEG raw recording session contained an average of 900 trials for left, right, passive imagery, initial relaxation, and experimental end period by a pseudo-random generator (“3 separate interaction segments per recording session and each segment took 15 minutes with an average of 300 trials”).

Each recording session is segmented into left-hand trials, right-hand trials, and passive trials separately by using the trial based method. The non-trials EEG segments such as “initial relaxation and experimental end period” are removed from all recording sessions. All of these preprocessing steps are done by using the EEGLAB toolbox. It is an interactive MATLAB toolbox for processing electrophysiological data such as EEG, MEG, and many others.

3.3. Source Imaging

Source imaging also known as source estimation is the estimation of brain sources from the sensor recording. The main objective of this paper is to analyze the effective ROI and source estimation that can discriminate among contributed brain regions for the specific behavior. However, source estimation (inverse problem) cannot directly contribute from the recording (sensor data), it requires the forward solution in the database. The forward solution is the calculation of the electrical potentials obtained at EEG sensors from the different head tissues such as the skull, scalp, and brain. Figure 3 shows the “EEG forward and inverse concepts” and this paper used Brainstorm MATLAB toolbox for both forward solution and inverse solution.

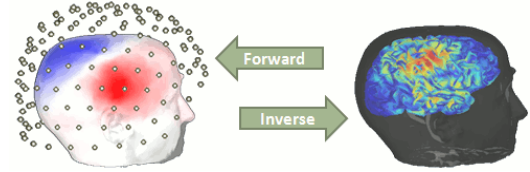


Figure 3: EEG forward and inverse concepts

As shown in Figure 4 (a), firstly, the preprocessed trial based segmented data is imported to Brainstorm toolbox for both forward and inverse solution. In this Figure 4 (a), “1,2, and 3” means “left hand, right hand, and passive/no imagery” events as detailed in section 3.1, and besides these events 321, 314, and 324 are the respective segmented trials for each event from the recording.

For the forward solution, this work used Brainstorm’s default anatomy, and it has the default source space of 15002 vertices as in Figure 4 (b). Among the available methods, unconstrained “OpenMEEG BEM” is used in this paper and it is the most effective method for the EEG forward problem. As a result, “OpenMEEG BEM” gives the “lead field matrix” [no of sensors * unconstrained no of sources] and that are the electric potentials from the different head tissues at the sensors.

Once the forward solution of head tissues and sensor characteristics are in the subject’s database, then we can reconstruct the brain sources that produced the respective sensors’ values. Weighted L2 minimum norm imaging (MNI) method is used to reconstruct sources (inverse solution) and it is the effective linear recombination by applying linear kernel. MNI gives the EEG “Imaging Kernel” matrix [no of vertices * no of channels]. Where the number vertices are perpendicular to the source space (15002 vertices) and the number of channels are 19 EEG channels.

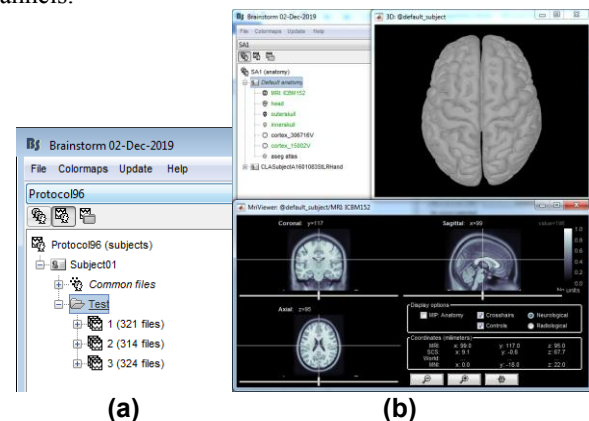


Figure 4: (a) Importing data to Brainstorm (b) Used default anatomy for the subjects

3.4. ROIs Analysis and Feature Extraction

MI EEG data does not include individuals' physical stimuli; it is only the mental task. In the literature, motor imagery EEG data is the rhythms of the sensory-motor cortex and relates to the frequency oscillation of alpha and beta bands.

The primary regions and parts of the brain are shown in Figure 5. In that, the sensory cortex and motor cortex are the central parts of the brain.

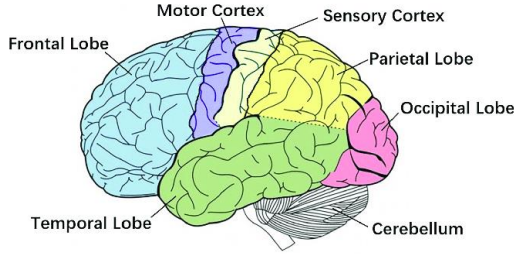


Figure 5: Primary regions/parts of the brain

Each of these primary parts in Figure 5: Frontal, Motor, Sensory, Parietal, Occipital, Cerebellum, and Temporal can be subdivided into many regions, so-called cortical parcellation or atlas. In the BCI jargon, a scout is a region of interest in the available source space and an atlas is a cortical parcellation with a list of scouts. The most commonly used cortical parcellations are Desikan-Killiany with 68 scouts and Destrieux Atlas with 148 scouts. Desikan-Killiany (DK) Atlas is used in this work. Table 1 and Figure 6 detailed DK brain structures' with subdivided regions and visualization of 68 scouts respectively. Figure 7 shows the examples of some analyzed ROIs based on Desikan-Killiany Atlas.

To analyze the robust ROI from Source Imaging, interested scouts are defined as the Regions of Interest. According to the literature, MI is the rhythms of the sensory-motor cortex. Firstly, this work defined an ROI with 4 regions near the sensory-motor cortex: such as "LC postcentral, RC postcentral, LC precentral, and RC precentral" (see in Figure 7).

Figure 8 shows the single-trial active responses for three different conditions from the defined 4 regions ROI. These responses represent the activity in specific brain regions and that are used as features to compare among different conditions: left hand, right hand, and passive imagery conditions. The SVM classifier is used to measure the effectiveness of the defined ROI responses between classes.

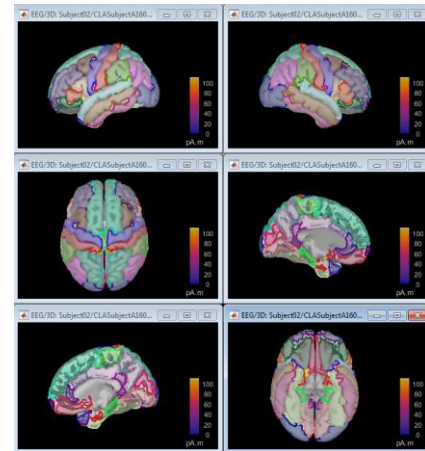
This work also defined and analyzed over 16 broad scouts that contained "RP supra marginal, LO lateral occipital, RO lateral occipital, LF superior frontal,

RF superior frontal, LC post central, RC post central, LC pre central, RC pre central, LP supra marginal, LT inferior temporal, RT inferior temporal, LT middle temporal, RT middle temporal, LT superior temporal, RT superior temporal".

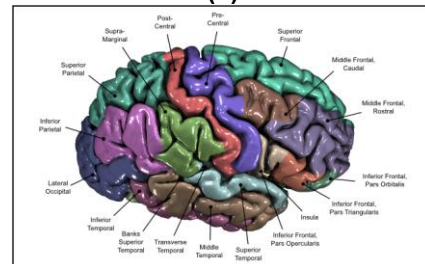
Multiple ROI analysis has been done over DK atlas and found a satisfaction result over 10 broad regions/scouts such as "RP supra marginal, LO lateral occipital, RO lateral occipital, LF superior frontal, RF superior frontal, LC postcentral, RC postcentral, LC precentral, RC precentral, LP supra marginal".

Table 1: Desikan-Killiany's brain structures

Regions	Desikan-Killiany (DK) atlas
Frontal	Caudal middle frontal, Frontal pole, Rostral middle frontal, Superior frontal, Lateral orbitofrontal, Medial orbitofrontal
Central	Para central, Post central, Pre central
Temporal	Inferior temporal, Middle temporal, Superior temporal, Temporal pole, Transverse temporal
Parietal	Parso percularis, Pericalcarine, Superior parietal, Supra marginal
Occipital	Lateral occipital, lingual pericalcarine



(a)



(b)

Figure 6: (a) Visualization Desikan-Killiany Atlas in Brainstorm (b) Desikan-Killiany Atlas with scouts labels

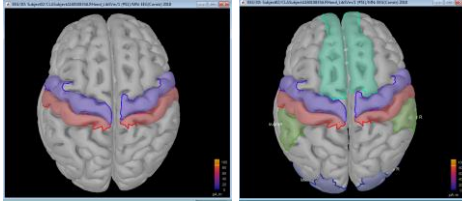


Figure 7: ROI extraction examples from Desikan-Killiany atlas

By classifying active responses of regions from analyzed ROIs, this work observed that not only the sensory-motor cortices but also the “superior frontal and lateral occipital” regions from both right and left cortices are effective regions for MI EEG data. The result and SVM classification results between ROIs are detailed in Figure 9.

4. Results and Discussions

This section shows the result among analyzed ROIs for the CLA paradigm (i.e. between three conditions: left hand, right hand, and passive/neutral). To measure the effectiveness of the analyzed ROIs, pairwise classification is done by using the LibSVM toolbox. The result and mean accuracy between classes for each ROI is shown in Figure 9.

In the classification process, 300 trials are used for each condition. Thus, for three conditions “left hand 300 trials, right hand 300 trials and passive 300 trials” are used from each subject. SVM performs classification separately (to discriminate between three classes) for each possible pair of classes: such as Right vs. Left, Passive vs. Left, and Passive vs. Right by using lower triangular elements in column-wise order function in MATLAB.

To reduce the computational load in the classification process, all trials are randomly assigned to 5 folds. All trials within each fold are subaverage into a single trial. Thus for 300 trials per class, a total of 5 subaverage trials yielding with 60 trials each for the recording are used to determine the defined ROI contains enough information for three classes.

According to the classification results in Figure 9, “10 scouts ROI” have high accuracy than “16 scouts ROI and “4 scouts ROI”. Also, “16 scouts ROI” are more reasonable for MI EEG data than only using the sensory motor cortex “4 scouts ROI”. Thus, the number of scouts in the ROI is not important for high accuracy. It depends on how each scout actively participants in a specific event or task.

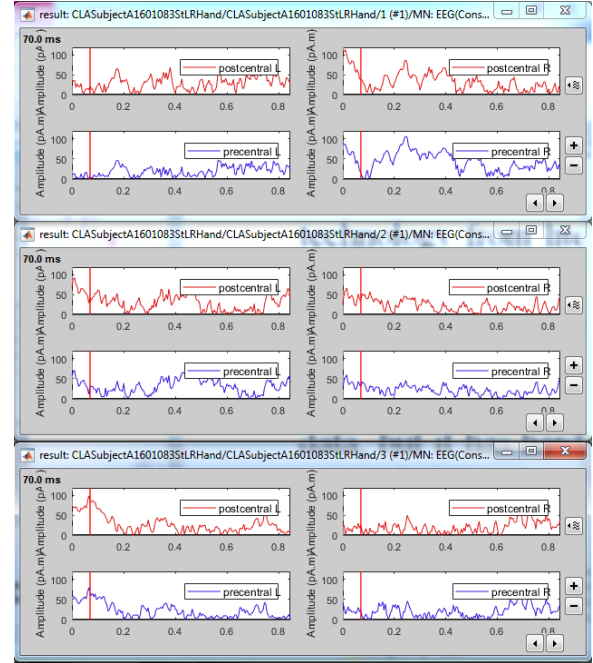


Figure 8: Single trial 4 regions ROI's active responses for three conditions (left hand, right hand, and passive)

5. Conclusions and Future

The main advantages of the EEG BCI system are easy to use, low costs, versatility, and portability of recent EEG devices. However, for the non-invasive brain imaging technology, it does not have a unique solution for source localization and ROIs estimation for the specific event or task. This paper combined preprocessing, segmentation, source imaging, ROIs analysis, and feature extraction. Moreover, these combine steps are executed with the most reliable methods in EEG BCI technology from literature. Finally, the SVM classifier is applied to measure the effectiveness of the proposed system, and accuracy between classes is shown in Figure 9.

Any special filters are not used on EEG raw data, but it has hardware filters by the collection system in the original data, detailed in section 3.2.1. To get a more classification accuracy and more consistency across subjects; in our future work, we will combine this system with some special noise filtering algorithms of EEG BCI system. And, deep learning is the most attractive way for machine learning, in the future work we will combine ROI imaging and deep learning technology, to discriminate effectively among MI EEG classes.

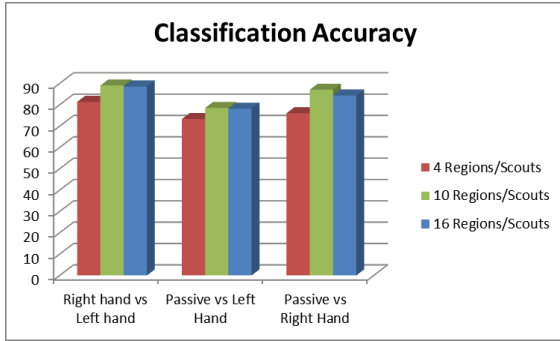


Figure 9: Classification results between classes with three different ROIs

6. Acknowledgement

The author would like to say deep thanks to the honorable adviser for their calm guidance and useful suggestions, also to the University of Information Technology for offering the required resources for running this research work. Finally, the author would like to give respect to the family for their great support and inspiration in all aspects of my work.

7. References

- [1] J. Müller-Gerking, G. Pfurtscheller, and H. Flyvbjerg, "Designing optimal spatial filters for single-trial EEG classification in a movement task," *Clin. Neurophysiol.*, vol. 110, no. 5, 1999, pp. 787–798.
- [2] N.Korhan, Z.Dokur, and T.Olmez, "Motor Imagery Based EEG Classification by Using Common Spatial Patterns and Convolutional Neural Networks", *Scientific Meeting on Electrical-Electronics & Biomedical Engineering and Computer Science (EBBT)*, 2019.
- [3] K.Mahjoory, V.Nikulim, L Botrel, K.Linkenkaer, M.Fato, and S.Haue, "Consistency of EEG source localization and connectivity estimates", *NeuroImage*.152, Jan 2017.
- [4] M.Xing, R.Tadayonnejad, A.MacNamara, O.Affilore, K.Luan, H.Klumpp, and A.Leow, "EEG Based Functional Connectivity Reflects Cognitive Load during Emotion Regulation", *IEEE 13th International Symposium on Biomedical Imaging (ISBI)*, 2016.
- [5] K. Murat, K. Mustafa, O. Erkan, Y. Hilmi and M. Yuriy "A large electroencephalographic motor imagery dataset for electroencephalographic brain computer interfaces", *Open Access*, 2018.
- [6] D. Christophe, F. Bruce, D. Anders, and H. Eric, "Automatic parcellation of human cortical gyri and sulci using standard anatomical nomenclature", *HHS Public Access*, 2011.
- [7] John C. Mosher, Richard M. Leahy,* and Paul S. Lewis, "EEG and MEG: Forward Solutions for Inverse Methods", *IEEE Transactions on Biomedical Engineering*, Vol. 46, No. 3, March 1999.
- [8] Gramfort et al, "OpenMEEG: opensource software for quasistatic Bioelectromagnetics", *BioMedical Engineering OnLine*, 2010.